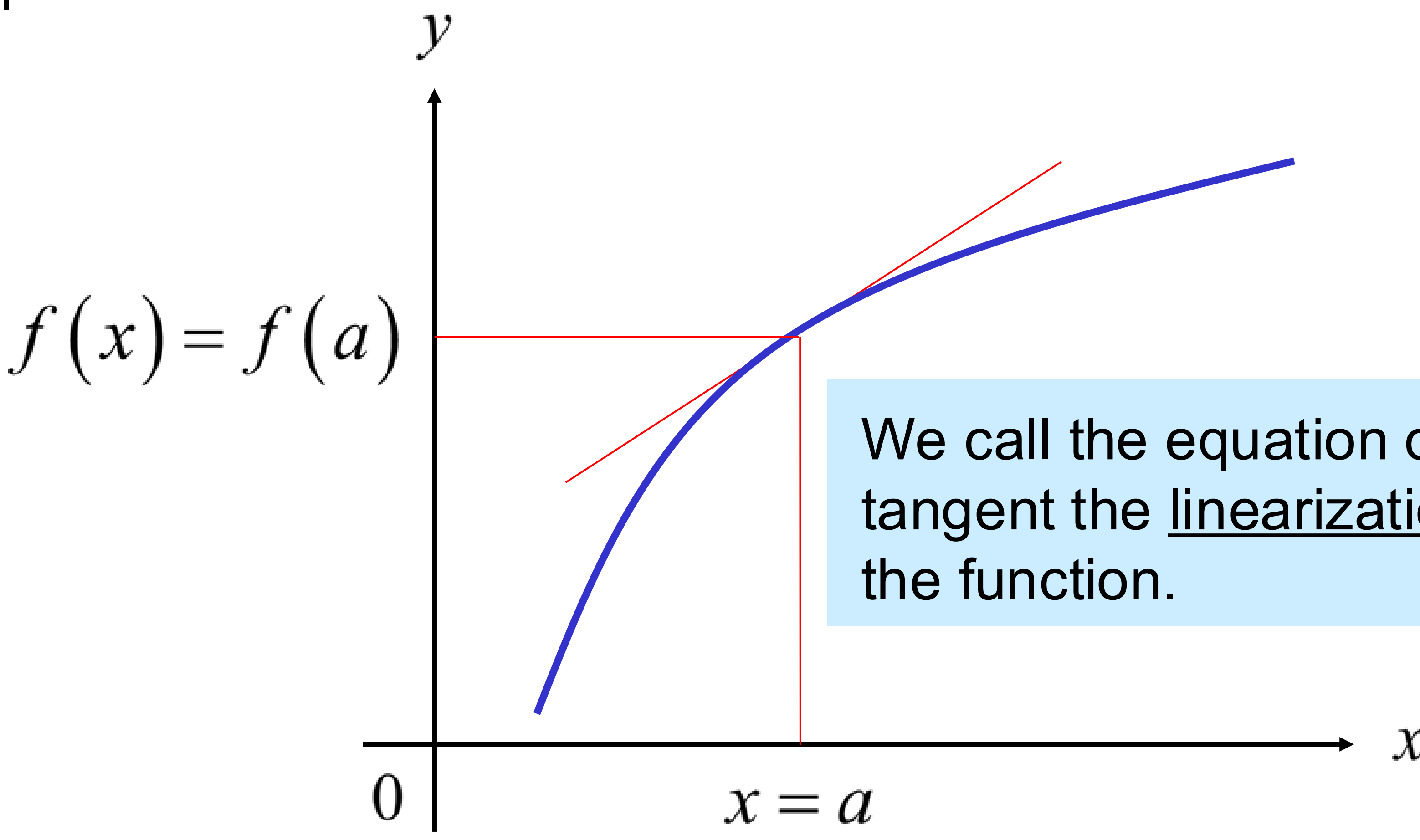
Linear Approximations, Differentials and Newton's Method

For any function $f(x)$, the tangent is a close approximation of the function for some small distance from the tangent point.



We call the equation of the tangent the <u>linearization</u> of the function.

Start with the point/slope equation:

$$y - y_1 = m(x - x_1) \qquad x_1 = a \qquad y_1 = f(a) \qquad m = f'(a)$$

$$y - f(a) = f'(a)(x - a)$$

$$y = f(a) + f'(a)(x - a)$$

$$\boxed{L(x) = f(a) + f'(a)(x - a)}$$

linearization of f at a

$f(x) \approx L(x)$ is the standard linear approximation of f at a.

The linearization is the equation of the tangent line, and you can use the old formulas if you like.

Important linearizations for x near zero:

$f(x)$	$L(x)$
$(1+x)^k$	$1+kx$
$\sin x$	x
$\cos x$	1
$\tan x$	x

Differentials:

When we first started to talk about derivatives, we said that $\frac{\Delta y}{\Delta x}$ becomes $\frac{dy}{dx}$ when the change in x and change in y become very small.

dy can be considered a very small change in y.

dx can be considered a very small change in x.

Let $y = f(x)$ be a differentiable function.

The differential dx is an independent variable.

The differential dy is: $dy = f'(x)dx$

Example: Consider a circle of radius 10. If the radius increases by 0.1, approximately how much will the area change?

$$A = \pi r^2$$

$$\frac{dA}{\cancel{dx}} = 2\pi r \frac{dr}{\cancel{dx}}$$

$$dA = 2\pi r \, dr$$

very small change in r

very small change in A

$$dA = 2 \cdot \pi \cdot 10 \cdot (0.1)$$

$$dA = 2\pi$$ (approximate change in area)

$$dA = 2\pi$$ (approximate change in area)

Compare to actual change:

New area: $\pi(10.1)^2 = 102.01\pi$

Old area: $\pi(10)^2 = \underline{100.00\pi}$

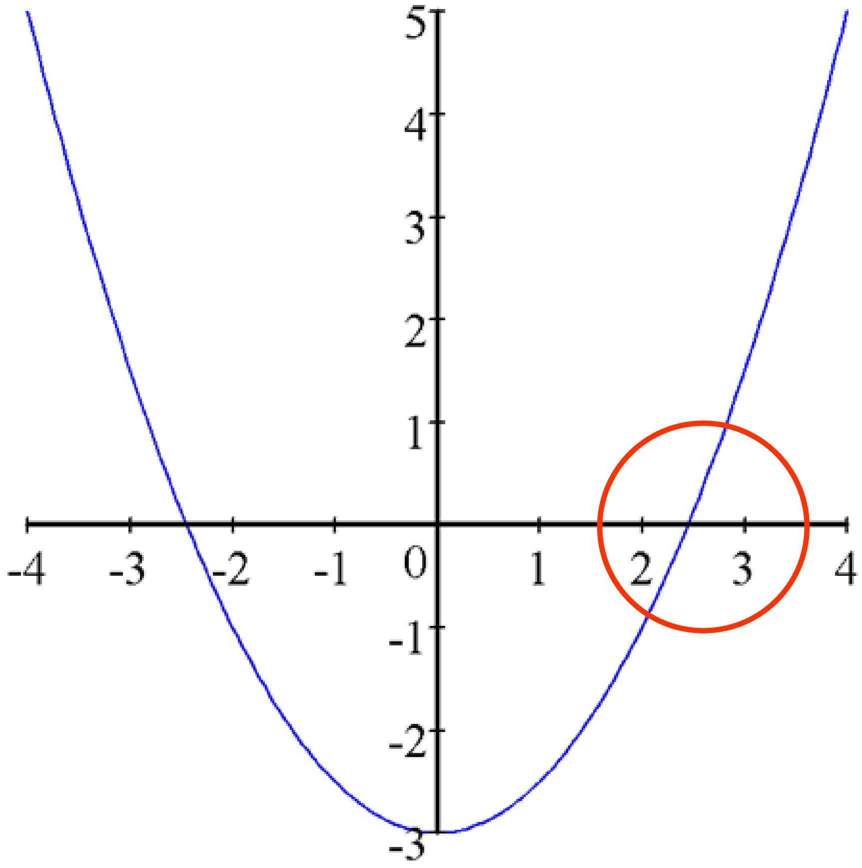
$$2.01\pi$$

$$\frac{\text{Error}}{\text{Actual Answer}} = \frac{.01\pi}{2.01\pi} \approx .0049751 \approx 0.5\%$$

$$\frac{\text{Error}}{\text{Original Area}} = \frac{.01\pi}{100\pi} \approx .0001 \approx 0.01\%$$

Newton's Method

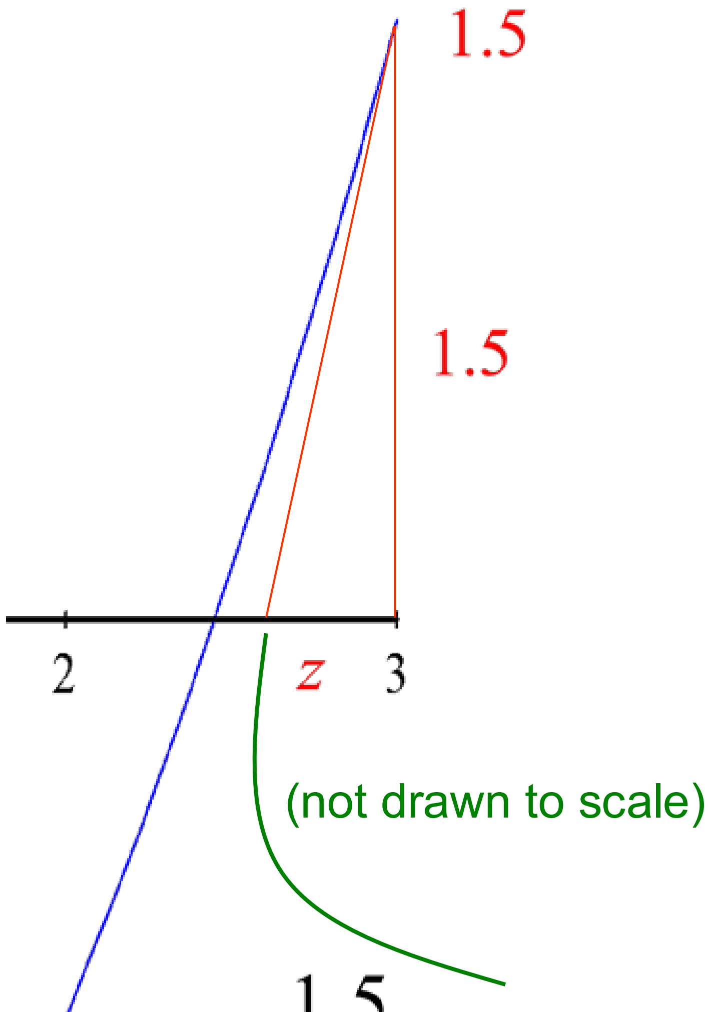
Finding a root for: $f(x) = \frac{1}{2}x^2 - 3$



We will use Newton's Method to find the root between 2 and 3.

$$f(x) = \frac{1}{2}x^2 - 3$$

$$f'(x) = x$$



Guess: 3

$$f(3) = \frac{1}{2} \cdot 3^2 - 3 = 1.5$$

$$m_{\text{tangent}} = f'(3) = 3$$

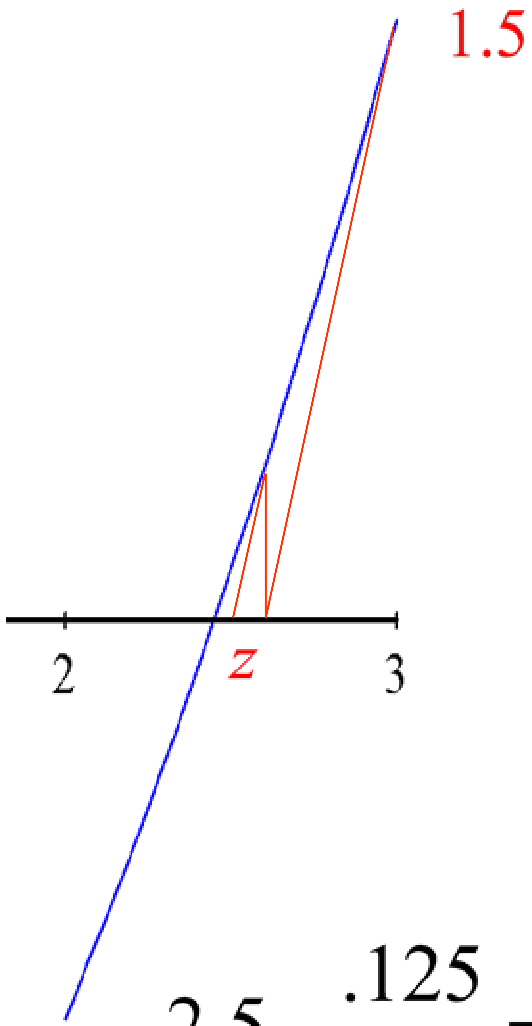
$$3 = \frac{1.5}{z} \qquad z = \frac{1.5}{3}$$

$$3 - \frac{1.5}{3} = 2.5$$

(new guess)

$$f(x) = \frac{1}{2}x^2 - 3$$

$$f'(x) = x$$



Guess: 2.5

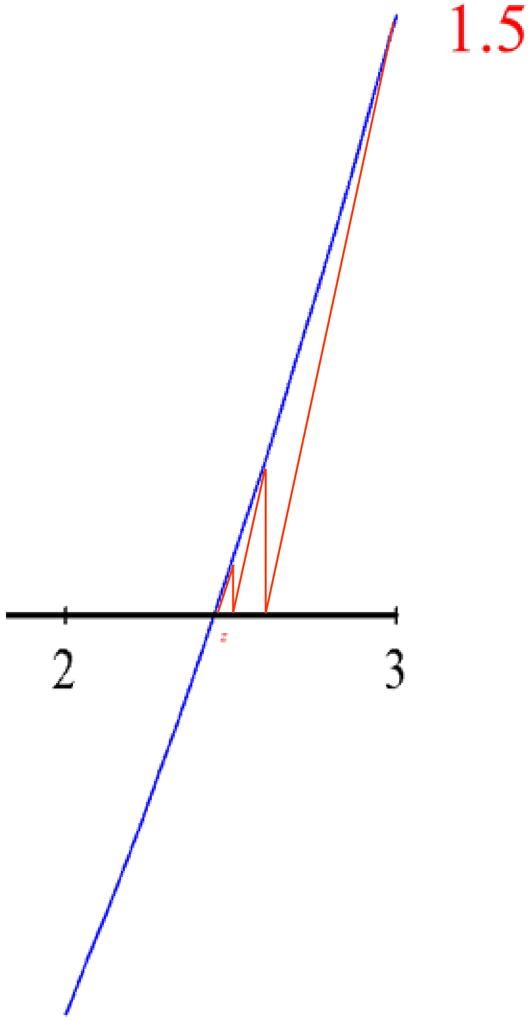
$$f(2.5) = \frac{1}{2}(2.5)^2 - 3 = .125$$

$$m_{\text{tangent}} = f'(2.5) = 2.5$$

$$z = \frac{.125}{2.5}$$

$$2.5 - \frac{.125}{2.5} = 2.45$$

(new guess)



$$f(x) = \frac{1}{2}x^2 - 3$$

$$f'(x) = x$$

Guess: 2.45

$$f(2.45) = .00125$$

$$m_{\text{tangent}} = f'(2.45) = 2.45$$

$$z = \frac{.00125}{2.45}$$

$$2.45 - \frac{.00125}{2.45} = 2.44948979592$$ (new guess)

Guess: 2.44948979592

$$f\left(2.44948979592\right) = .00000013016$$

Amazingly close to zero!

This is Newton's Method of finding roots. It is an example of an algorithm (a specific set of computational steps.)

This is a recursive algorithm because a set of steps are repeated with the previous answer put in the next repetition. Each repetition is called an iteration.

Newton's Method: $$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

This is Newton's Method of finding roots. It is an example of an algorithm (a specific set of computational steps.)

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This technique is sometimes called the Newton-Raphson method.

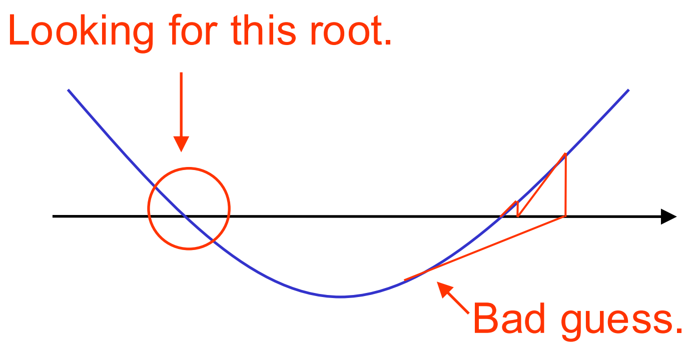
Find where $y = x^3 - x$ crosses $y = 1$.

$1 = x^3 - x \quad 0 = x^3 - x - 1 \quad f(x) = x^3 - x - 1 \quad f'(x) = 3x^2 - 1$

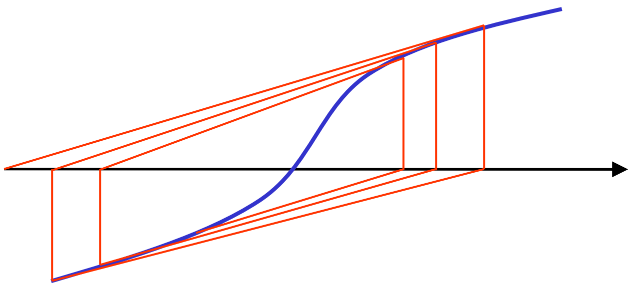
n	x_n	$f(x_n)$	$f'(x_n)$	$x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$
0	1	-1	2	$1 - \dfrac{-1}{2} = 1.5$
1	1.5	.875	5.75	$1.5 - \dfrac{.875}{5.75} = 1.3478261$
2	1.3478261	.1006822	4.4499055	1.3252004

$$(1.3252004)^3 - 1.3252004 = 1.0020584 \approx 1$$

There are some limitations to Newton's method:



Wrong root found



Failure to converge